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Access the code, data, and analysis at https://github.com/eteitelbaum/code-satp.git

The Limits and Promise of Automated Event Coding*

Evidence from the South Asia Terrorism Portal

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ABSTRACT

This paper evaluates the extent to which modern transformer-based language models can viably substitute for human coders in applied conflict research. Using hand-coded data from the South Asia Terrorism Portal on the Maoist insurgency in India, we assess contemporary NLP approaches with attention to the key challenges that arise in real-world conflict event coding. We evaluate six encoder-based transformer models across three common classification tasks and compare their performance to few-shot generative baselines from the GPT family. We find that automated coding performs well on frequent and lexically distinctive categories even with limited training data, while more complex and context-dependent labels remain challenging. Domain-specific pretraining offers advantages in low-data regimes and for semantically subtle categories, but strong general-purpose encoders often achieve comparable performance as labeled data increase. Baseline models fail on the rarest categories, but targeted imbalance handling strategies substantially improve performance without degrading accuracy on common labels. Taken together, the results suggest that machine coding can meaningfully complement and, in some settings, approximate human annotation, while also clarifying the architectural choices and methodological steps required to maximize the practical value of transformer models in conflict research.

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Introduction

The systematic coding of political events from textual sources has long been a bottleneck for conflict research. Major datasets like ACLED and UCDP-GED rely on large teams of trained human analysts (Eck 2012), while smaller projects depend on research assistants whose funding is often temporary. This reliance on labor-intensive coding has repeatedly motivated efforts to automate event extraction. Early automation efforts using dictionary-based systems like KEDS, TABARI, and GDELT achieved impressive scale but struggled with paraphrase, implicit references, and contextual reasoning. The more recent emergence of transformer-based language models has fundamentally transformed this landscape, with evaluations showing that both fine-tuned encoder models and large generative models can match or exceed human coding quality on a range of political text classification tasks (Gilardi, Alizadeh, and Kubli 2023; Törnberg 2025; Ornstein, Blasingame, and Truscott 2023; Heseltine and Clemm von Hohenberg 2024). These advances have raised the prospect that automated systems might soon complement or replace human coders at scale, but how close are we to that reality, and what unresolved questions stand in the way?

We argue that for conflict studies, the core question is not whether automation is possible, but which approaches are reliable under the conditions that real-world datasets present. We highlight and explore three unresolved issues: whether fine-tuned encoders or large generative decoders are better suited to structured event coding; whether rare but analytically central categories can be recovered when training data are sparse; and the extent to which domain-specific pretraining significantly improves performance over strong general-purpose models. These debates have direct consequences for scholars deciding how to build, scale, and validate conflict event data.

We investigate these questions through a systematic evaluation of modern NLP tools for conflict event classification, using hand-coded data from the South Asia Terrorism Portal (SATP) on the Maoist insurgency in India. We evaluate six encoder-based transformer models (BERT, DistilBERT, ELECTRA, RoBERTa, XLNet, and the domain-specific ConfliBERT) alongside decoder-only baselines from the GPT family across three classification tasks: perpetrator identification, action type coding, and target type coding. Our focus is on classification rather than other extraction tasks such as named entity recognition or question-answering-based approaches to event coding (Zhong, Dhuliawala, and Stoehr 2023; Halterman et al. 2021), though the tasks we examine capture core dimensions of structured political event annotation. The dataset reflects the kind of bespoke coding common in applied conflict research, with labels that vary in frequency and complexity and include niche categories tailored to the specific analytic interests of the research team. We examine how performance varies with training data availability and inference speed, and systematically evaluate six imbalance handling strategies for rare categories.

Our results suggest that the relative merits of encoder and decoder architectures depend on the nature of the task. Fine-tuned encoders surpass a GPT-4.1 mini few-shot baseline at modest training fractions for perpetrator classification and target type coding, but the generative baseline performs surprisingly well on action types, where categories like "bombing" and "armed assault" are lexically distinctive and described in standardized language. Target types, by contrast, are more context-dependent and often described using local terminology, including regional institutional designations and conflict-specific group identifiers, where a general-purpose model has less exposure. This pattern cuts against the claim that generative models hold a blanket advantage on complex classification.

On the domain-specificity question, we find that ConfliBERT performs well, particularly in low-data regimes and on semantically subtle labels, but general-purpose RoBERTa matches its performance on many tasks at comparable inference speed. Overall, the evidence points to a more nuanced picture. Domain-specific pretraining offers clear advantages in some settings, but it does not confer an unconditional edge across all tasks. With sufficient labeled data, strong general-purpose encoders can be competitive.

Baseline encoder models fail entirely on the rarest categories, but targeted imbalance handling strategies, particularly back-translation augmentation through contextually relevant languages, achieve substantial improvements on previously unlearnable labels. Even so, some rare categories remain stubbornly difficult, pointing to limits of current techniques when training examples are extremely scarce. Taken together, these findings offer practical guidance for conflict researchers navigating a rapidly evolving NLP toolkit, whether working with limited computational resources, bespoke annotation schemes, or imbalanced label distributions. The optimal approach depends on the availability of labeled data, the lexical complexity of the coding scheme, and the frequency distribution of the categories involved.

The remainder of the paper proceeds as follows. We first review the literature on NLP approaches to conflict event coding, discussing theoretical expectations for encoder versus decoder architectures, the challenge of rare categories, and the role of domain-specific pretraining. We then describe the SATP dataset and experimental design, present results across three classification tasks, and evaluate imbalance handling strategies. We conclude with implications for researchers seeking to automate conflict event coding.

NLP Approaches to Conflict Event Coding

Automated event coding has evolved through three distinct generations. Dictionary-based systems like KEDS, TABARI, and their successors matched text against hand-built pattern dictionaries tied to the CAMEO ontology, enabling projects like GDELT and ICEWS to process millions of news reports at unprecedented scale (Schrodt 2012). However, these systems were brittle in that they struggled with implicit references and paraphrasing, and required constant manual maintenance as topics and domain-specific language evolved. Supervised machine learning classifiers improved flexibility by learning patterns from labeled examples rather than dictionaries (Croicu and Weidmann 2015; Halterman et al. 2023), while hybrid pipelines like SPEED combined automated pre-filtering with human review (Nardulli, Althaus, and Hayes 2015). The arrival of transformer-based language models marked a qualitative break, offering the ability to learn contextual representations of text that generalize across phrasing and sentence structure (Terechshenko et al. 2020; Widmann and Wich 2023; Li et al. 2024). Today, the question is no longer whether automated coding is feasible but which architectural approach best suits which coding task.

Encoders and Decoders: Two Logics of Classification

Transformer models used for text classification generally fall into two broad families that embody distinct logics. Encoder-only models like BERT and its variants process the full input simultaneously using bidirectional self-attention, generating contextual representations that capture how each token relates to every other token in the sequence (Devlin et al. 2019). This architecture is optimized for supervised classification, learning to associate input patterns with output labels through fine-tuning on human-coded examples. The resulting classifier operates through what might be called associative logic, learning stable statistical regularities that map input text to labels within the training distribution. In practice, this approach typically requires a substantial number of labeled examples for fine-tuning to perform well, particularly for complex or fine-grained classification tasks (Zhang et al. 2021; Mosbach, Andriushchenko, and Klakow 2021).

Decoder-only models like ChatGPT or Claude process text autoregressively, generating output one token at a time conditioned on all preceding tokens (Brown et al. 2020). When applied to classification, these models translate the task into a text generation problem: given a prompt describing the coding scheme and one or more examples, the model produces a label as natural language output. This architecture brings a different kind of capability. Because large generative models are trained on vast, multi-domain corpora, they carry implicit world knowledge that allows them to make approximate inferences (or "reason") about references not explicitly stated in the input text (Törnberg 2025; Ornstein, Blasingame, and Truscott 2023). A generative model can potentially infer that a report mentioning "red corridor operations" describes a Maoist action even without having seen that phrase paired with the Maoist label during training, because it has encountered the phrase in other contexts during pretraining. As a result, decoder-only models are often applied in zero-shot or few-shot settings, relying on prompt specification rather than task-specific fine-tuning.

These differences suggest practical choices for conflict event coding. Encoder-based classification excels when the mapping between textual features and labels is consistent and well-represented in training data. When reports describing bombings reliably contain words like "blast," "IED," or "detonated," a fine-tuned encoder will learn these associations quickly and apply them reliably; but when categories require interpretation of implicit actors, indirect references, or contextually dependent language, the associative approach may fall short. The abstraction gap between text-level features and the higher-order social science constructs that event coding schemes require is precisely where generative models might hold an advantage.

Yet the empirical evidence regarding the advantages of decoder-only models is mixed. Generative models perform well on binary or coarse-grained classification tasks but remain unstable on fine-grained event coding with many categories (Hu et al. 2024a). Few-shot prompting is sensitive to example selection and prompt formulation, and the nondeterminism inherent in autoregressive generation complicates reproducibility (Abdurahman et al. 2025). Zero-shot relation classification approaches also show promise in political texts (Hu et al. 2024b). Fine-tuned encoders, meanwhile, offer decisive advantages in throughput and cost and run on standard academic computing resources rather than requiring expensive API access (Brandt et al. 2025). For projects coding tens of thousands of events, these considerations may weigh as heavily as marginal differences in accuracy.

The relative merits of the two approaches may therefore be task-specific, depending on the interaction between task characteristics and data availability. We expect fine-tuned encoders to dominate on well-represented categories where consistent textual patterns exist in the training data, particularly once the training set exceeds the roughly 1,000 labeled examples at which supervised encoder models have been shown to overtake few-shot generative baselines in downstream classification accuracy (Pangakis and Wolken 2024). Generative models should be most competitive on categories that are rare in the training data or that require contextual reasoning beyond surface-level lexical cues. The central question that we investigate is where the crossover point between the two sets of models lies, and where it falls for different classification tasks.

Rare Categories and the Limits of Supervised Learning

Class imbalance is endemic to conflict event data. Political violence datasets are dominated by a small number of common event types, while the categories that are often most analytically interesting, such as abductions, targeted assassinations, or attacks on specific infrastructure, occur infrequently. This creates a fundamental problem for supervised learning: models trained on imbalanced data learn to predict majority classes and neglect minority ones, producing classifiers that appear accurate in aggregate but fail on exactly the labels researchers care about most.

The challenge is compounded by what Simon et al. (2025) call the abstraction gap. Some categories are lexically distinctive. For example, reports of bombings almost always mention explosions, devices, or detonations. Other categories require higher-order inference. An "abduction" might be described as someone being "taken away," "picked up by a group," or simply "disappeared," with no single lexical signature. When a category is both rare and lexically diffuse, standard supervised models lack both the training signal and the textual anchors needed to learn it reliably.

The machine learning literature offers several families of approaches to class imbalance. Loss function modifications such as class-weighted cross-entropy and focal loss increase the penalty for misclassifying minority examples, effectively forcing the model to attend to rare categories during training. Sampling strategies including random oversampling and weighted sampling alter the composition of training batches to expose the model to more minority examples. Data augmentation techniques generate synthetic training examples for underrepresented classes, with approaches ranging from simple back-translation through contextually relevant languages (Halterman 2025) to paraphrase generation using domain-specific models like Confli-T5 (Parolin et al. 2022). More recent work has explored knowledge distillation, using a large generative model to produce training labels for rare categories that can then be used to augment the training data for smaller, faster classifiers (Pangakis and Wolken 2024).

These techniques are well-established in machine learning but remain underexplored in the conflict event coding literature. We systematically evaluate six imbalance handling strategies across our classification tasks. We expect that techniques operating at the data level, particularly augmentation through back-translation and paraphrase generation, will outperform loss-level modifications on the rarest categories, because they address the underlying scarcity of training signal rather than simply reweighting existing examples. However, we also expect that some categories will remain stubbornly difficult regardless of technique, those that combine extreme rarity with low lexical distinctiveness, pointing to limits of what current supervised methods can achieve.

Domain-Specific Pretraining: How Much Does It Help?

A parallel line of research has invested in building transformer models pretrained on conflict-specific corpora. The most prominent of these is ConfliBERT, a BERT-base model pretrained on a large collection of political violence texts including news reports, academic articles, and NGO documentation (Hu et al. 2022). The intuition is straightforward: a model that has already encountered conflict-specific vocabulary, entity names, and narrative structures during pretraining should require less fine-tuning data to achieve strong performance on downstream conflict coding tasks. Related efforts include Multi-CoPED and 3M-Transformers, which extend domain-specific pretraining to multilingual conflict corpora (Skorupa Parolin et al. 2022; Parolin et al. 2021).

The expected advantage of domain-specific pretraining is clearest in low-data regimes. When fine-tuning data is scarce, a model that already understands conflict terminology starts from a better position than one pretrained on general-purpose text. Domain-specific models may also handle rare entities and specialized vocabulary more effectively, since terms like “Salwa Judum” or “CRPF” that appear frequently in conflict reporting but rarely in general English corpora will be better represented in the model’s vocabulary and contextual embeddings.

But the magnitude of any domain-specific advantage is an empirical question that depends on how much fine-tuning data is available. With sufficient labeled examples, general-purpose models can learn task-specific patterns that compensate for the absence of domain-specific pretraining. At the same time, not all encoder architectures are equal. For example, even within the BERT family of models, newer models like RoBERTa can outperform the original BERT on standard conflict-related classification tasks, likely because of differences in pretraining data and objectives (Liu et al. 2019). We therefore expect ConfliBERT’s advantages to be most visible at small training fractions and on labels involving conflict-specific terminology, and to diminish as fine-tuning data increases and stronger general-purpose encoders have enough examples to learn the relevant patterns directly.

The SATP Maoist Violence Incidents Dataset

The South Asia Terrorism Portal (SATP) is among the most comprehensive open-source repositories of conflict data on South Asia, offering curated incident summaries that span decades of political violence across the subcontinent. For researchers studying India's Maoist (Naxalite) insurgency, SATP has been an indispensable resource. Economists have used it to study the effects of workfare programs on insurgent violence (Fetzer 2020),[1] to identify the economic determinants of the conflict (Ghatak and Eynde 2017), to trace how insurgents select targets in response to state capacity (Vanden Eynde 2018), and to examine the fiscal incentives shaping counterinsurgency operations (Shapiro and Vanden Eynde 2023). In the NLP domain, Hu et al. (2022) used SATP data to evaluate ConfliBERT on binary and multi-label classification tasks, and Brandt et al. (2024) extended this comparison to decoder-based models.

Coding Ontology

Our dataset draws on the same underlying SATP incident reports but imposes a bespoke coding ontology tailored to the dynamics of the Maoist conflict. The most prominent existing ontology for terrorism event data is that of the Global Terrorism Database (GTD), which categorizes incidents along dimensions including attack type (9 categories, from assassination to unarmed assault), target/victim type (22 categories with further subtypes), weapon type (13 categories), and perpetrator group (LaFree and Dugan 2007). The GTD ontology is designed for global coverage of terrorism broadly defined and is correspondingly expansive (START 2021). Its coding rules reflect the goal of cataloguing diverse forms of political violence across hundreds of conflicts and countries.

Our coding scheme departs from the GTD framework in ways that reflect both the structure of the Maoist conflict and our analytical priorities. We code three perpetrator classes: Maoist insurgents, Security Forces, and Other. This is simpler than the GTD's perpetrator fields, which accommodate up to three named groups per incident with confirmed/unconfirmed designations, but it captures the essential actor distinction in a two-sided insurgency where the identity of the acting party is often the first question analysts need answered.

For action types we code seven categories: Armed Assault, Bombing, Abduction, Seizure, Arrest, Surrender, and Facility/Infrastructure Attack. Several of these map onto GTD attack types (Armed Assault, Bombing/Explosion, Hostage Taking, Facility/Infrastructure Attack), but arrests, seizures and surrenders have no GTD equivalents because they describe non-violent state or insurgent actions rather than terrorist attacks.

For target types we code nine categories: Security, Civilians, Government Officials, Government Infrastructure, Private Property, Mining Companies, NGOs, Other Groups, and Maoists.

[1] Fetzer's work also involved an early effort to automate SATP coding using SENNA, a dictionary-based NLP toolkit. The limitations of that approach, touched on earlier, motivate the transformer-based methods we evaluate here.

This is a deliberate compression of the GTD's 22 target types, collapsing distinctions (such as between airports, maritime targets, and transportation) that are irrelevant to a land-based rural insurgency while preserving those (such as between mining companies and other private property) that matter for understanding Maoist targeting patterns.

Lexical Complexity

The coding categories in our ontology vary considerably in their lexical distinctiveness, a property that has direct implications for classification difficulty. Some categories are associated with predictable, concentrated vocabulary. Bombing incidents, for example, reliably feature terms like *landmine*, *blast*, *IED*, *detonated*, and *explosion*. Consider the following incident summary:

> CPI-Maoist cadres trigger a series of landmine blasts targeting a Police party combing the forests of Murmur village in the Wajeedu area of Khammam District.

This incident is coded as Perpetrator: Maoist, Action: Bombing, Target: Security. The key terms map straightforwardly to the coding categories, and a dictionary-based system could plausibly identify the action type from keyword matching alone.

Other categories are lexically diffuse, meaning that they can be described using varied and overlapping vocabulary. Armed Assault encompasses ambushes, firefights, shootings, and stabbings, with vocabulary that overlaps substantially with other action types. Perpetrator identification often depends on contextual inference rather than explicit mention. For example:

> A 35-year-old man, believed to be a SPO of Chhattisgarh Police, was stabbed to death by suspected militia members of the CPI-Moist during a cockfighting event near Kommanapalli village in Dummugudem mandal in Khammam District of Andhra Pradesh.

This incident is coded as Perpetrator: Maoist, Action: Armed Assault, Target: Security. But classifying it correctly requires knowing that SPO refers to Special Police Officer (a target type inference), recognizing "suspected militia members of the CPI-Moist" as a Maoist attribution despite the hedging language, and categorizing a stabbing as an armed assault rather than some other action type. A dictionary-based system would need entries for every such variant whereas a transformer model can potentially learn these associations from context. The variation in lexical distinctiveness across our coding categories provides a natural testing ground for several of the expectations developed earlier, namely that the relative performance of encoders and decoders will depend on the interaction between category frequency and lexical complexity, and that imbalance handling strategies will matter most for low-frequency and lexically diffuse categories.

Dataset Construction

Our dataset spans 2005 to 2016, a period selected to align with the peak intensity of Maoist activity across central and eastern India. It comprises approximately 10,000 incident reports and was constructed through a multi-year coding project involving undergraduate coders trained in political science and South Asian studies, working under the supervision of graduate researchers with domain expertise. Coders relied on a detailed codebook and entered structured data using a standardized digital form. Each report was parsed to determine whether it described a single event or multiple events, and every incident received a unique identifier encoding state, date, and event sequence.

Quality control was a central concern throughout the coding process. Coders underwent structured onboarding, participated in supervised trial phases, and engaged in regular adjudication sessions to resolve ambiguous cases. A significant share of the dataset was double-coded, with discrepancies reviewed and resolved by senior coders. When coding standards evolved, previously coded events were revisited to harmonize labels across the full time series. The resulting dataset supports three classification tasks: a multi-class perpetrator task (3 classes), a multi-label action type task (7 categories), and a multi-label target type task (9 categories).

Methods

To evaluate how effectively pretrained language models can recover structured political violence data from unstructured text, we implemented a series of classification experiments using our hand-coded SATP Maoist Violence Incidents Dataset. We evaluate model performance on three supervised learning tasks corresponding to key dimensions of political violence. The first is a multi-class perpetrator task in which models predict which of three actor types (Maoist, Security, or Other) is responsible for a given incident. The second and third are multi-label tasks in which models predict one or more action types (7 categories) and target types (9 categories), respectively. The specific label sets for each task are detailed in the results sections below.

Encoder Models

The primary focus of this study is on fine-tuned encoder-only transformer models, which generate bidirectional contextual embeddings and are well suited to classification tasks that require attending to all parts of the input simultaneously. We selected six models for evaluation, all freely available via the Hugging Face Model Hub:

- *BERT-base-cased* (Devlin et al. 2019): A 12-layer, 110M parameter model pretrained on BooksCorpus and Wikipedia. It serves as the baseline encoder model for classification tasks.

- *DistilBERT-base-cased* (Sanh et al. 2020): A 6-layer, 66M parameter distilled version of BERT that is faster and lighter, designed to reduce inference cost with minimal performance loss.
- *RoBERTa-base* (Liu et al. 2019): A 12-layer, 125M parameter BERT variant trained with more data and no next sentence prediction, generally outperforming BERT on downstream tasks.
- *ELECTRA-base-discriminator* (Clark et al. 2020): A 12-layer, 110M parameter model that uses a more sample-efficient pretraining objective (replaced token detection) rather than masked language modeling.
- *XLNet-base-cased* (Yang et al. 2020): A 12-layer, 110M parameter autoregressive model trained with permutation-based language modeling, capable of modeling bidirectional context while retaining autoregressive benefits.
- *ConfliBERT* (Hu et al. 2022): A BERT-base model pretrained on a corpus of political violence and conflict-related text. It is domain-specific and designed to improve performance on tasks involving political conflict data.

All six models fall into the "base" category (between 66–125M parameters), making them feasible to run on commodity hardware, including laptops and free or low-tier Google Colab environments. This makes them particularly well suited for use in low-resource settings, both in terms of computational infrastructure and access to large labeled datasets. Their relative efficiency also allows for more flexible experimentation, such as evaluating performance across multiple training data fractions.

Training Procedure

Each model was fine-tuned on a fixed training–validation–test split drawn from the full dataset. Approximately 2,000 incident descriptions were randomly selected and held out at the start: 1,000 for validation and 1,000 for final testing. The remaining ~8,000 incidents were used in the training loop, where we iteratively varied the size of the training set across six fractions: 1/32, 1/16, 1/8, 1/4, 1/2, and the full training set. For the multi-label tasks, we used iterative stratification to preserve label co-occurrence distributions across splits (Sechidis, Tsoumakas, and Vlahavas 2011; Szymański and Kajdanowicz 2017). Validation and test sets remained fixed throughout, allowing consistent comparison of performance across data fractions and model architectures. All models were fine-tuned using their cased versions, based on the assumption that capitalization conveys important semantic and contextual information in this domain, particularly for identifying proper nouns, such as location names, political figures, or militant organizations, which are common in South Asia–specific reporting.

Fine-tuning used the AdamW optimizer with a learning rate of 2×10^{-5}, weight decay of 0.01, and a batch size of 32 for both training and evaluation. Models were trained for 2 epochs with evaluation at the end of each epoch, and the best checkpoint was selected based on validation micro F1. Input sequences were tokenized with a maximum length of 128 tokens for the perpetrator task and 512 tokens for the multi-label tasks, with padding and truncation applied uniformly. All experiments used a fixed random seed of 42 across Python, NumPy, and PyTorch to ensure reproducibility.

Decoder Baseline

To contextualize the encoder results, we also evaluated three decoder-only LLMs from the GPT family in both zero-shot and few-shot settings: GPT-4o-mini, GPT-4.1 mini, and GPT-5.2. These models were accessed through the OpenAI API and evaluated on the same held-out test sets used for the encoder models. In the few-shot condition, each prompt included a small number of labeled examples (three for perpetrator, five for action type and target type) drawn from outside the test set. The system prompt provided category definitions, and the model was instructed to return only the predicted label(s). Full prompts are reproduced in Section . We set temperature to 0 to maximize determinism. Response parsing used fuzzy string matching against the label set, with unmatched outputs mapped to a default category.

Among the three decoder models, GPT-4.1 mini few-shot achieved the best overall balance of cost and accuracy and is used as the baseline reference throughout the results. Performance varied little across the three models (see Section); we therefore do not treat decoder model selection as a major source of variation.

Evaluation Metrics

To evaluate model performance, we report weighted macro F1 scores for the multi-class perpetrator classification task and micro F1 scores for the two multi-label classification tasks (action type and target type). Weighted macro F1 accounts for class imbalance by weighting each class's F1 by its support, while micro F1 aggregates true positives, false positives, and false negatives across all labels before computing F1, making it appropriate for multi-label settings where label frequency varies widely. Confidence intervals are obtained via bootstrap resampling of the test set predictions. Additional performance metrics, including precision, recall, and per-label F1 scores, are provided in an online supplement.

Imbalance Handling Strategies

Because several categories in the action type and target type tasks are rare (e.g., Mining Company appears in fewer than 1% of incidents), baseline models often fail to learn reliable decision boundaries for these labels. To isolate the effect of imbalance handling from model architecture, we evaluated six strategies applied to a single architecture (ConfliBERT): per-label threshold tuning, focal loss, conservative class weighting, weighted random sampling, back-translation augmentation through contextually relevant languages (Hindi, Urdu, Bengali), and T5 paraphrase augmentation. These strategies span three broad approaches (modifying the loss function, rebalancing the training distribution, and augmenting the training data) and are described in detail alongside their results in a later section.

Results

Perpetrator Task

The perpetrator task is a multi-class classification problem in which models are trained to predict the actor type responsible for a given incident, with classes including Maoist, Security, and Other. Figure 1a displays the weighted macro F1 scores achieved by each model across different training data fractions. Figure 1b focuses on a single fraction (1/8 of the training set), presenting 95% confidence intervals obtained via bootstrap resampling (5,000 iterations) to assess statistical uncertainty around model estimates.

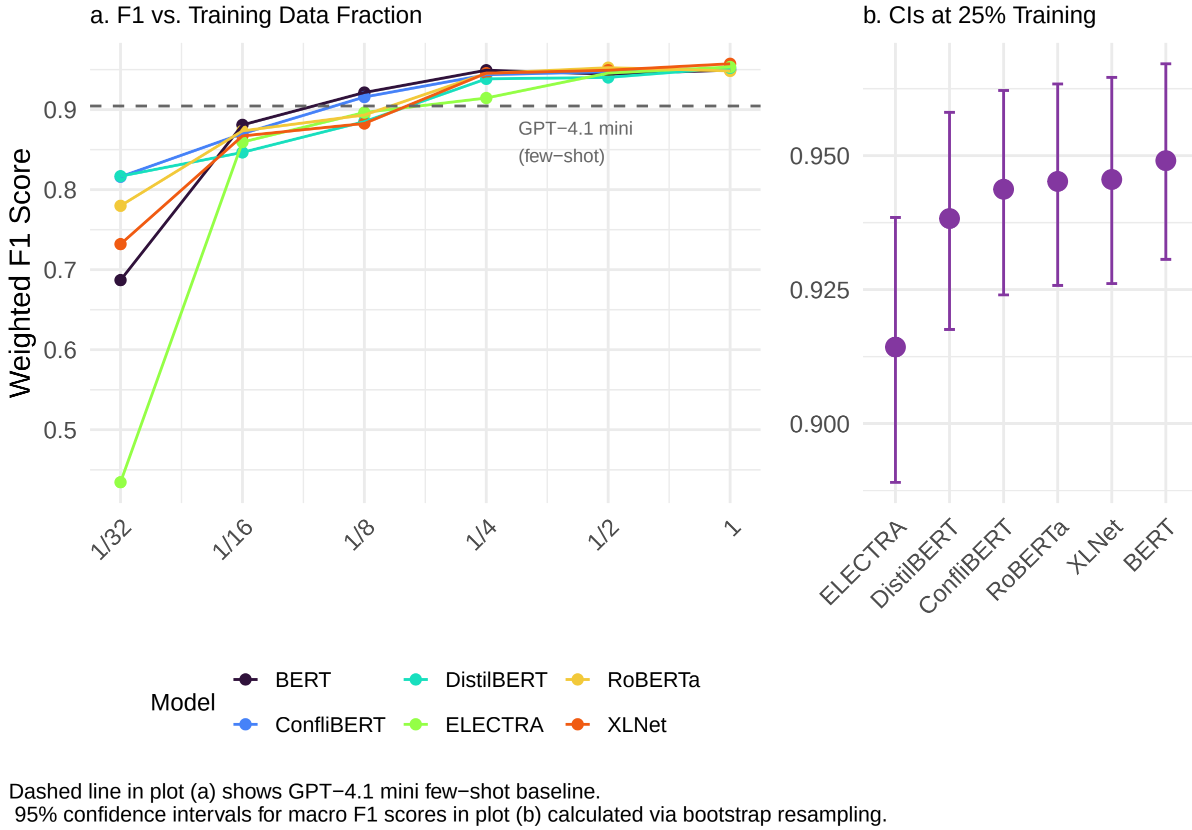


Figure 1: Model Performance for 'Perpetrator' Task

The dashed line in Figure 1a marks the GPT-4.1 mini few-shot baseline, which achieves a weighted F1 of 0.905 without any task-specific fine-tuning. Every encoder model surpasses this level once trained on roughly one-eighth of the labeled data, illustrating how quickly fine-tuning closes and then widens the gap relative to prompting alone.

Consistent with findings reported in the original ConfliBERT paper (Hu et al. 2022), we observe that ConfliBERT performs particularly well at lower data fractions, where its domain-specific pretraining appears to confer an advantage. However, the gap narrows quickly. At the 1/8 training level, RoBERTa and XLNet both achieve F1 scores that are not statistically distinguishable from ConfliBERT's, perhaps complicating a straightforward narrative about domain-specific model superiority (Brandt et al. 2024; Osorio et al. 2024).

When trained on the full dataset, all models converge to very high weighted macro F1 scores, ranging from 0.957 (ConfliBERT) to 0.968 (RoBERTa). This tight clustering in performance likely reflects the relative simplicity of the classification task (predicting whether an actor is Maoist, Security, or Other) and the benefit of high-quality, consistent human annotations. While ConfliB-

ERT retains a slight advantage in low-resource settings, the results at full training suggest that general-purpose encoder models are capable of matching or exceeding the performance of domain-specific alternatives on this task. In a later section, we return to inference speed and other performance trade-offs, which further contextualize these results for applications in low-resource environments.

Action Type Task

The second task is a multi-label classification problem in which models predict one or more action types associated with a given incident. These include Armed Assault, Bombing, Abduction, Seizure, Arrest, Surrender, and Facility/Infrastructure Attack. Performance is evaluated using micro F1 scores, which are better suited to imbalanced, multi-label settings where many samples carry multiple labels or none at all.

Figure 2a presents model performance across increasing training data fractions, with the GPT-4.1 mini few-shot baseline shown as a dashed line at 0.896. The decoder-only baseline performs well on this task, achieving a competitive score without any fine-tuning. While encoders do surpass it with sufficient training data, the margin at full training is modest; the best encoder (XLNet, 0.939) exceeds the GPT baseline by only a few points, suggesting that action type categories are distinctive enough for a general-purpose model to classify from examples alone. Most models cross the GPT baseline by the 12.5% training fraction. As in the perpetrator task, we observe notable variation in learning rates. RoBERTa and XLNet clearly lead at the 25% training level (Figure 2b), with micro F1 scores of 0.882 and 0.879, respectively. ConfliBERT trails modestly behind at 0.857, while BERT and DistilBERT cluster around 0.800 and 0.785. ELECTRA performs substantially worse, achieving a score of only 0.636 at this level.

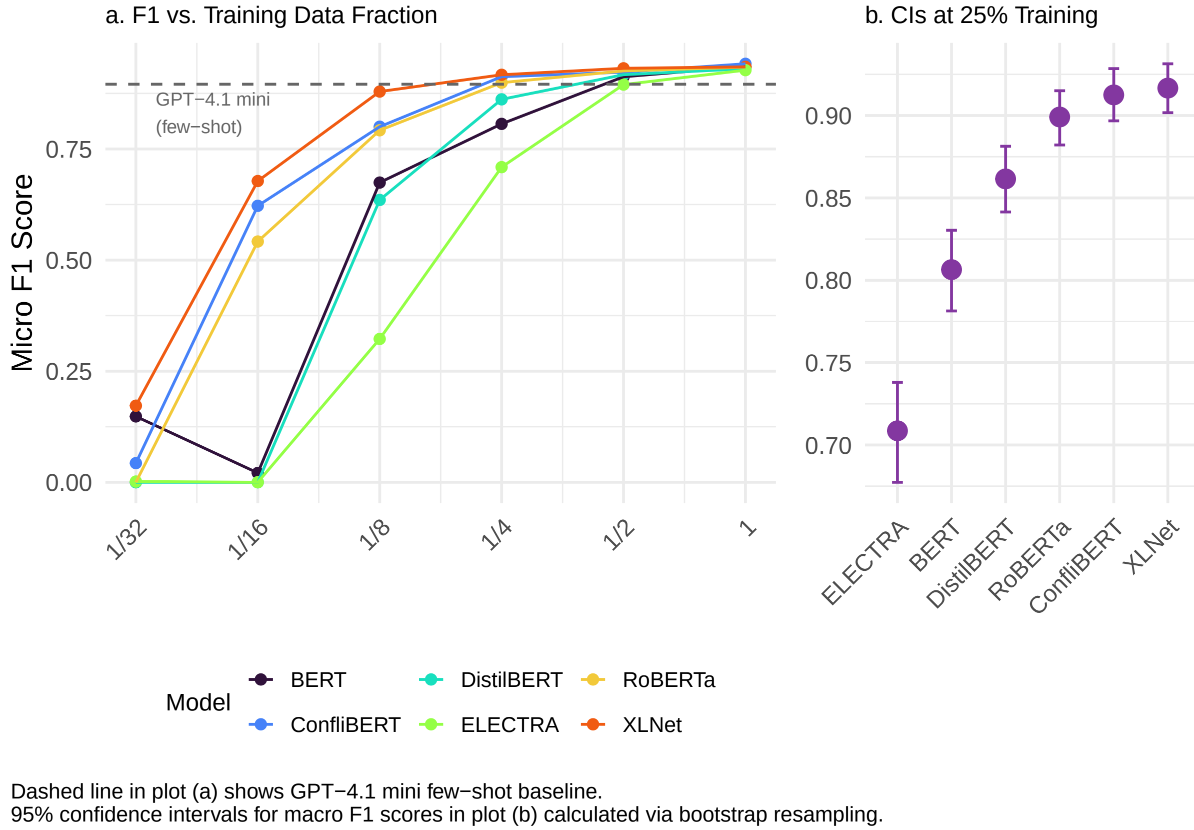
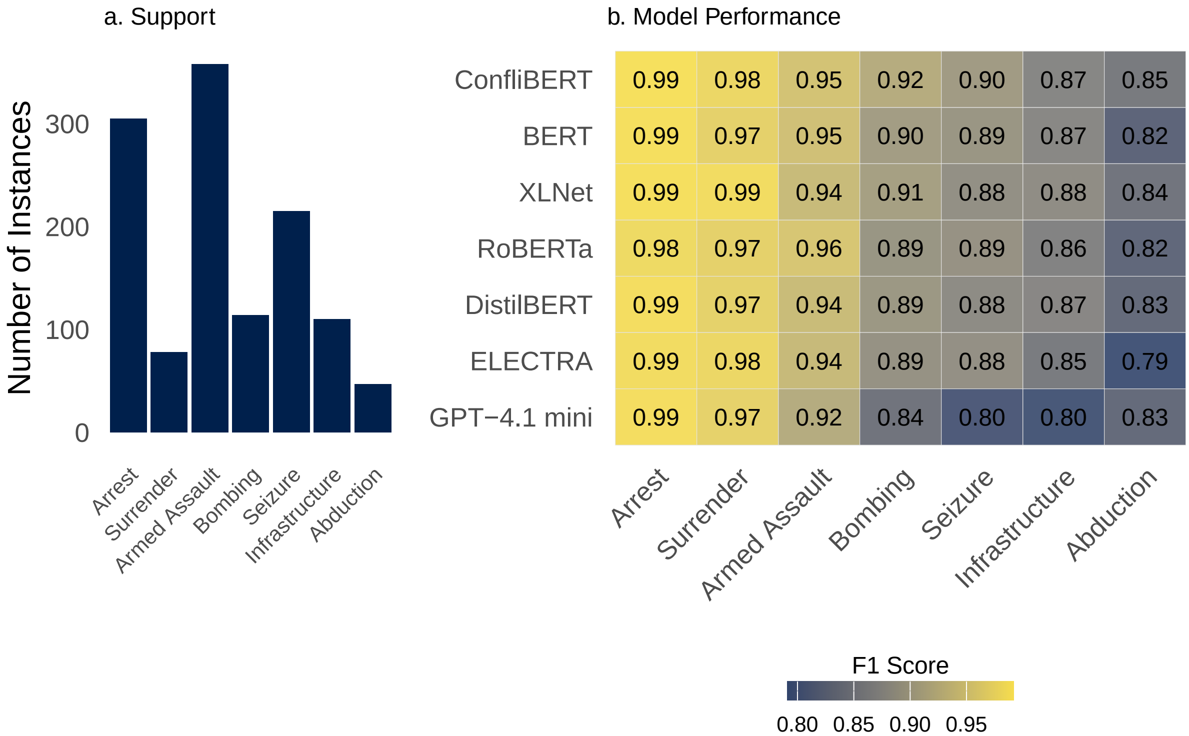


Figure 2: Model Performance for 'Action Type' Task

When trained on the full dataset, all models perform well, converging to micro F1 scores in a narrow band between 0.915 (ELECTRA) and 0.939 (XLNet). The top four models (XLNet, RoBERTa, BERT, and ConfliBERT) are separated by only a few hundredths of a point, underscoring a high level of overall convergence. Once again, ConfliBERT performs strongly in low-data regimes but is closely matched or outperformed by general-purpose encoder models when more training data are available.



Barplot in plot (a) shows the number of samples per label in the test set.
Heatmap in plot (b) shows the F1 score for each model and label.

Figure 3: Model Performance by 'Action Type' Label

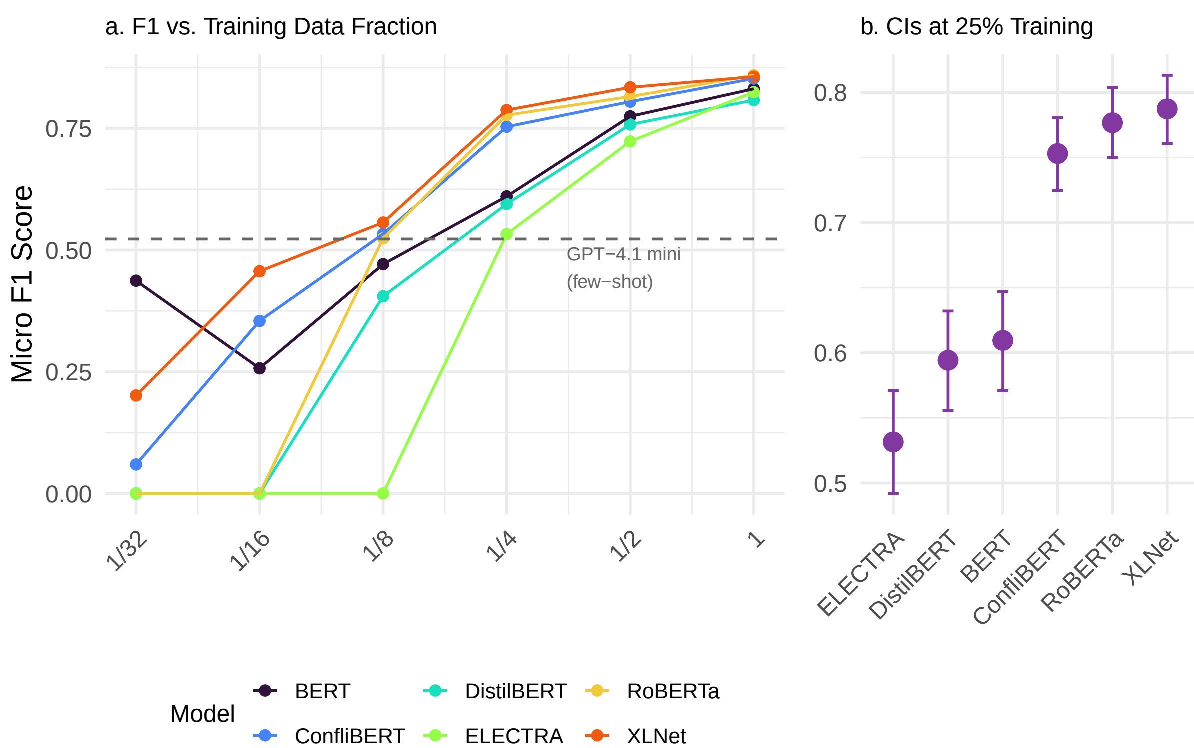
These headline results, however, mask important variation across action types. As shown in Figure 3, which includes GPT-4.1 mini alongside the encoder models, some labels (e.g., Surrender) are predicted with very high accuracy despite low support in the training set, while others (e.g., Facility/Infrastructure Attack) perform less well despite being relatively more common. The GPT baseline shows a similar pattern: it handles lexically distinctive categories well but struggles with labels that require more nuanced contextual understanding. This suggests that label frequency alone does not fully explain model performance. Certain labels may benefit from linguistic distinctiveness. For example, the word "surrender" appears with few synonyms or paraphrases in the source texts, while others, like infrastructure-related actions, may involve greater lexical ambiguity or more diverse expressions, making them harder to classify.

Target Type Task

The final task involves predicting one or more target types associated with each incident. As in the action classification task, this is a multi-label setting, and performance is again evaluated using micro F1 scores. Target categories include Security, Civilians, Officials, Government

Infrastructure, Private Property, Mining Companies, NGOs, Other Groups, and Maoists.

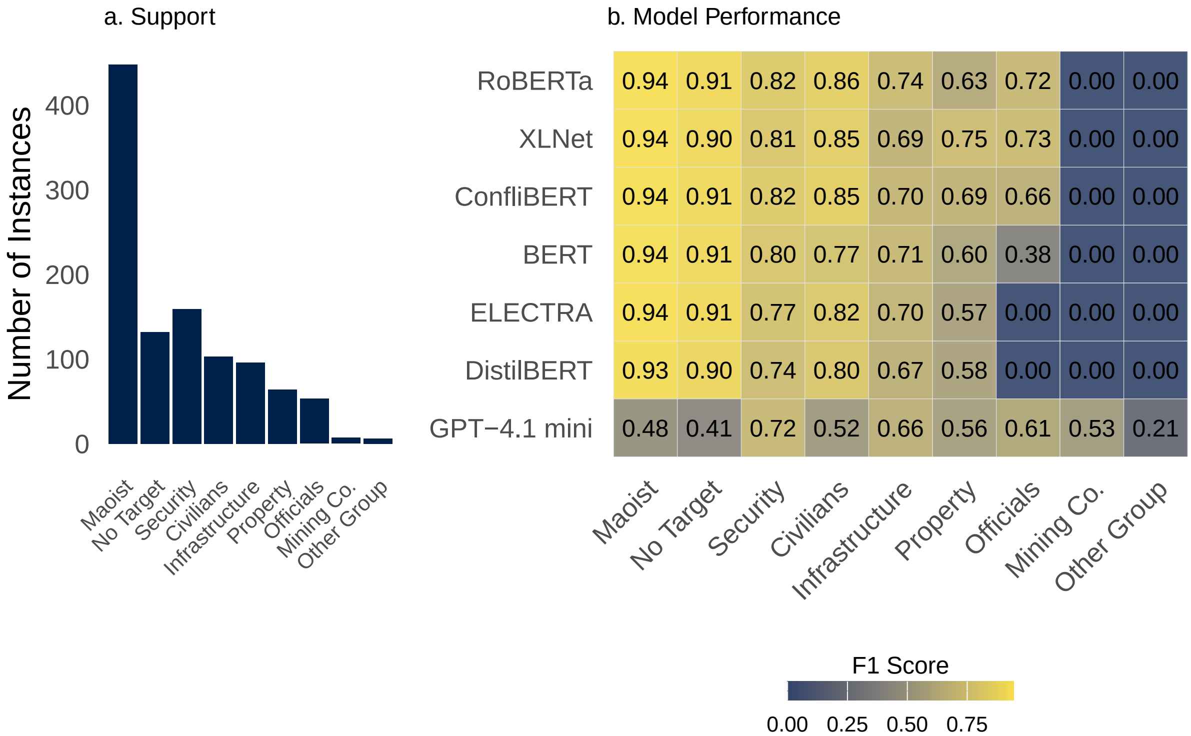
Figure 4a presents model performance across training fractions. The GPT-4.1 mini few-shot baseline (dashed line at 0.523) is notably lower here than on the other two tasks, reflecting the greater difficulty of target type classification for a model relying solely on in-context examples. Figure 4b zooms into the 50% training level, where we again observe a clear performance cluster among XLNet, ConfliBERT, and RoBERTa. At this level, XLNet leads with a micro F1 score of 0.839, followed by ConfliBERT (0.819) and RoBERTa (0.816). DistilBERT and BERT score considerably lower (0.770 and 0.710, respectively), and ELECTRA again falls well behind the others at 0.614.



Dashed line in plot (a) shows GPT−4.1 mini few−shot baseline.
95% confidence intervals for macro F1 scores in plot (b) calculated via bootstrap resampling.

Figure 4: Model Performance for 'Target Type' Task

At full training (Figure 4a, rightmost points), model scores converge somewhat, but still lag those observed in the action type task. ConfliBERT reaches 0.878, with RoBERTa (0.870) and XLNet (0.869) close behind. However, performance tails off for BERT (0.847), DistilBERT (0.826), and ELECTRA (0.787). These figures indicate that target prediction is a more difficult task overall, likely due to higher class imbalance and greater semantic ambiguity in the expression of target entities.



Barplot in plot (a) shows the number of samples per label in the test set.
Heatmap in plot (b) shows the F1 score for each model and label.

Figure 5: Model Performance by 'Target Type' Label

Figure 5 further illustrates this point by breaking down F1 scores across individual labels. Some target types, particularly those with limited support in the training set, show markedly lower performance. However, the decline is not solely attributable to frequency. For example, Surrender in the action task achieved strong scores despite low support, likely due to its lexical distinctiveness. By contrast, Infrastructure/Facility targets underperform even when relatively frequent, suggesting that greater lexical variability and ambiguity in how these targets are described contributes to model error.

Performance-Speed Tradeoffs

Figure 6 presents a performance–speed comparison for the action and target type classification tasks, plotting micro F1 scores against evaluation speed (measured in samples per second). The figure highlights the trade-offs between model accuracy and runtime efficiency, which are especially relevant for researchers working in low-resource or latency-sensitive environments.

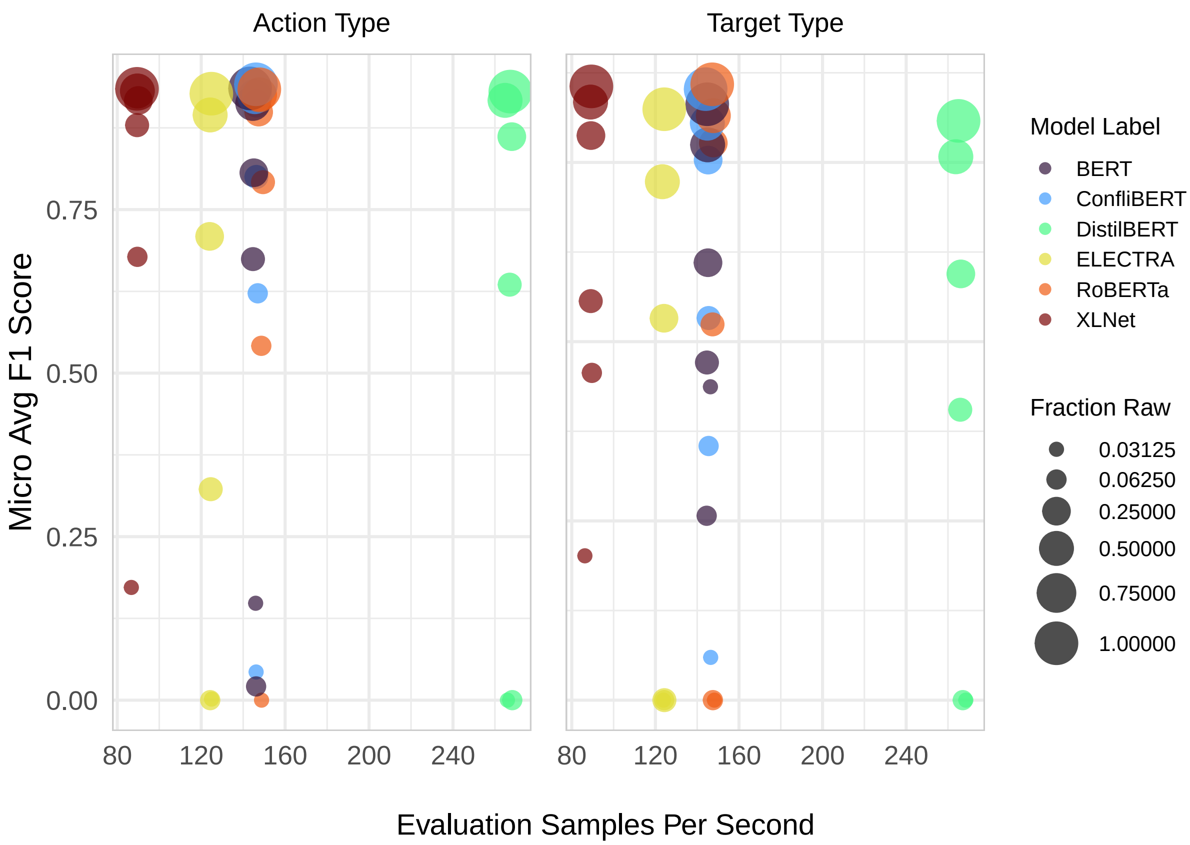


Figure 6: Model Performance vs. Speed for Multiclass Classification Tasks

Among the top-performing models, RoBERTa and ConfliBERT stand out for their efficiency. At full training, both achieve micro F1 scores near the top of the range (0.870 and 0.878, respectively, on the target task) while maintaining high evaluation speeds of 147 and 145 samples per second. BERT, while slightly less performant (0.847), operates at a comparable speed (144 samples/sec), making it a viable option where pretraining data diversity is sufficient.

In contrast, XLNet, though highly competitive in terms of accuracy (0.869), is much slower, evaluating at only 65 samples per second, less than half the speed of RoBERTa and ConfliBERT. This substantial performance-cost trade-off limits XLNet's practical appeal in time-constrained settings.

DistilBERT emerges as the fastest model, running at 267 samples per second, but as seen earlier, it tends to underperform at lower training fractions, making it less suitable for small-data scenarios. ELECTRA performs poorly on both axes: it is the least accurate model (0.787) and also slower than most of its peers (124 samples/sec), suggesting limited utility for either high-accuracy or speed-critical tasks.

Though Figure 6 focuses on the action and target type tasks, patterns are similar for the perpetrator classification task (not shown). In all cases, the results suggest that RoBERTa and ConfliBERT offer the best balance between accuracy and efficiency, particularly when high throughput is required alongside strong classification performance.

Improving Low-Frequency Label Performance

While our baseline encoder models demonstrated strong performance on frequent target types such as "Maoist," "Security," and "Civilians", they exhibited complete failure on the rarest categories. Specifically, models achieved zero F1 scores for "Mining Company" and "Non-Maoist Armed Group" across all six encoder architectures tested. This pattern reflects the common challenge in conflict event classification in that some of the most operationally significant events such as those involving corporate targets or non-state armed actors often represent the smallest fractions of training.

This class imbalance pattern, while not extreme, creates meaningful challenges for classification performance. In our SATP training dataset, "Mining Company" targets comprise 54 instances (0.7% of training examples) while "Non-Maoist Armed Group" targets appear in 48 instances (0.6% of training examples), compared to 3,583 "Maoist" examples (45.1%) and 1,270 "Civilian" examples (16.0%). Such low-frequency labels create a challenging learning environment where standard loss functions and sampling procedures struggle to develop reliable decision boundaries.

Imbalance Handling Strategies

To systematically address these challenges, we implemented six distinct imbalance handling strategies using ConfliBERT as our base model, each targeting different aspects of the class imbalance problem through modifications to loss functions, sampling procedures, or training data augmentation. We selected ConfliBERT for this analysis given its strong performance across the classification tasks explored earlier in this paper and its domain-specific pretraining on conflict-related texts.

Baseline Strategy. Our baseline strategy provides a controlled comparison using standard ConfliBERT training procedures with binary cross-entropy loss and default 0.5 prediction thresholds. This baseline leverages ConfliBERT's domain-specific representations while employing conventional training approaches that struggle with class imbalance. All experiments employed comprehensive reproducibility controls, including fixed random seeds (42) across Python, NumPy, PyTorch, and CUDA operations to ensure reliable comparisons across strategies.

Threshold Tuning. We tested threshold tuning both as a standalone imbalance handling strategy and integrated it into the five subsequent strategies explored below. As a standalone strategy, threshold tuning represents a computationally efficient strategy requiring no model retraining and therefore serves as an important first-line approach for addressing class imbalance in operational settings. We use macro PR-AUC for checkpoint selection during training to ensure good representations across all classes before applying post-training calibration and threshold optimization. The threshold optimization process then starts with temperature scaling applied to model outputs to improve probability calibration (Guo et al. 2017), followed by grid search optimization across threshold values from 0.05 to 0.95 (19 points) to maximize micro-F1 performance on the validation set.

Class Weights. This traditional approach employs inverse frequency weighting within the loss function, computing weight = $N_{\text{negative}}/N_{\text{positive}}$ for each label and applying these weights via PyTorch's BCEWithLogitsLoss (He and Garcia 2009). This conceptually simply method provides a strong baseline for comparison with more sophisticated approaches.

Weighted Sampling. The weighted sampling strategy balances training batches through strategic oversampling of underrepresented examples using PyTorch's WeightedRandomSampler (Buda, Maki, and Mazurowski 2018). Sample weights are computed as 1 / (label_count + 1), where label_count represents the sum of positive labels per example. This ensures that rare examples appear multiple times per epoch while enabling replacement sampling to maintain batch diversity.

Focal Loss. The focal loss strategy addresses class imbalance through a modified loss function that dynamically focuses learning on hard-to-classify examples (Lin et al. 2018). We replaced standard binary cross-entropy with focal loss using automatically computed per-label alpha weights:

$$\alpha_{\text{pos}} = \text{clamp}\left(\sqrt{\frac{\text{median prevalence}}{\text{label prevalence}}}, 0.25, 0.75\right)$$

where *median_prevalence* represents the typical frequency across all target types in our dataset, *label_prevalence* is the frequency of the specific target type being weighted, and *clamp* constrains the resulting weight to fall between 0.25 and 0.75 to ensure training stability. This formulation automatically assigns higher weights to rare labels like "Mining Company" (0.7% prevalence) while reducing weights for common labels like "Maoist" (45.1% prevalence). The gamma parameter was set to 2.0 for standard imbalanced labels and boosted to 2.5 for ultra-rare labels (prevalence $< 1\%$). This adaptive parameterization eliminates manual hyperparameter tuning while ensuring appropriate focus on the most challenging examples.

Back-Translation Augmentation. Our back-translation strategy generates synthetic training examples for underrepresented categories through translation-based paraphrasing (Sennrich, Haddow, and Birch 2016; Halterman 2025). We employed an automatic threshold-based approach, targeting all labels with fewer than 900 positive training examples to systematically address imbalance across "Mining Company" (54 training examples), "Non-Maoist Armed Group" (48 examples), "Private Property" (509 examples), and "Government Infrastructure" (771 examples). The strategy translates English incident summaries through Hindi, Urdu, and Bengali, languages contextually relevant to South Asian conflict data, before back-translating to English using Google Translate API. Synthetic generation was constrained by safety parameters: a maximum of 500 new examples per label and a synthesis ratio preventing synthetic examples from exceeding original positive counts. The strategy preferentially selects single-label positive examples to minimize label co-occurrence artifacts, effectively doubling training data for the most underrepresented categories while providing controlled expansion for moderately imbalanced labels.

T5 Paraphrase Augmentation. This strategy employs Google FLAN-T5-Large for high-quality paraphrase generation under identical threshold-based selection criteria and safety constraints as back-translation (Chung et al. 2022; Parolin et al. 2022). The approach uses structured prompts emphasizing factual preservation and neutral tone, with quality control through similarity filtering using sentence-transformers (all-MiniLM-L6-v2), accepting paraphrases with cosine similarity between 0.85-0.98 to original text. Generation parameters (temperature=0.7, top_p=0.9) ensure semantic consistency while creating meaningful variation. The same label selection and synthesis ratio constraints as back-translation are applied to achieve a balanced augmentation strategy that enhances representation for rare categories without overwhelming the original data distribution.

Improvements on Low-Frequency Categories

Our imbalance handling strategies achieved transformative improvements on previously unlearnable categories, as illustrated in Figure 7. For example, "Mining Company" targets improved from complete failure (0.00 F1) across all baseline models to substantial performance under weighted sampling (0.625 F1), while "Non-Maoist Armed Group" targets reached 0.857 F1 using back-translation augmentation, representing infinite relative improvement over baseline performance.

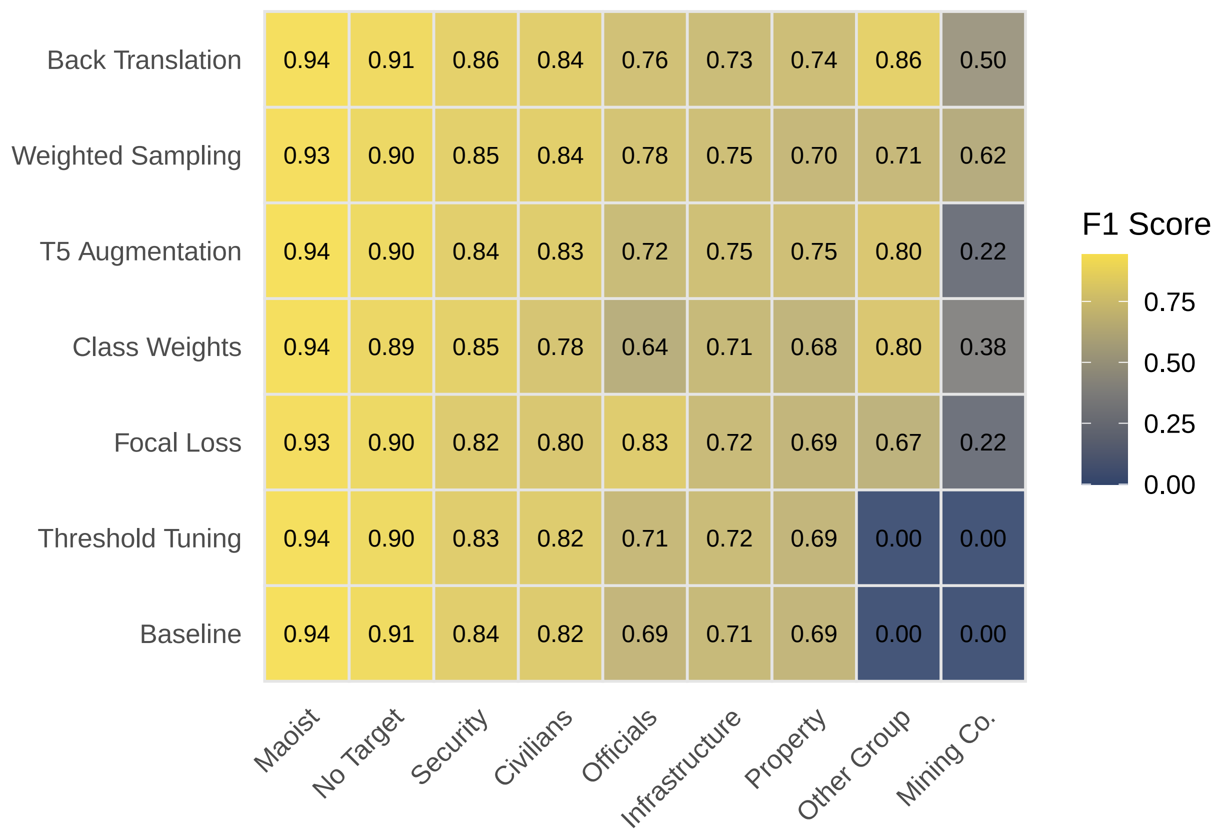


Figure 7: Performance of Six Imbalance Handling Strategies

The heatmap reveals distinct strategy profiles across the target type spectrum. Back translation augmentation emerged as the most consistently effective approach, achieving strong performance across both rare categories (Mining Company: 0.50, Non-Maoist Armed Group: 0.857). T5 augmentation demonstrated comparable effectiveness with particularly strong performance on "Government Infrastructure" (0.751) and "Private Property" (0.748) targets.

Weighted sampling showed remarkable effectiveness for "Mining Company" targets (0.625 F1) and solid performance on "Civilians" (0.842), suggesting that oversampling strategies may be particularly valuable for targets with moderate scarcity. Class weights and focal loss provided meaningful but more modest improvements, while threshold tuning alone, despite sophisticated optimization, failed to overcome the fundamental learning challenges posed by extreme class imbalance.

Crucially, these improvements on rare categories were attained without degrading performance on frequent targets. All strategies maintained strong F1 scores above 0.82 for "Security," "Civilians," and "Maoist" categories, indicating that the approaches successfully addressed imbalance without creating negative transfer effects.

Precision-Recall Tradeoffs in Rare Category Classification

The precision-recall plots for our four target rare categories (Figure 8) illuminate the distinct approaches each strategy takes to the precision-recall optimization problem. These plots reveal that achieving high performance on rare categories requires navigating fundamental tradeoffs between conservative precision and comprehensive recall, with different strategies occupying distinct regions of the precision-recall space.

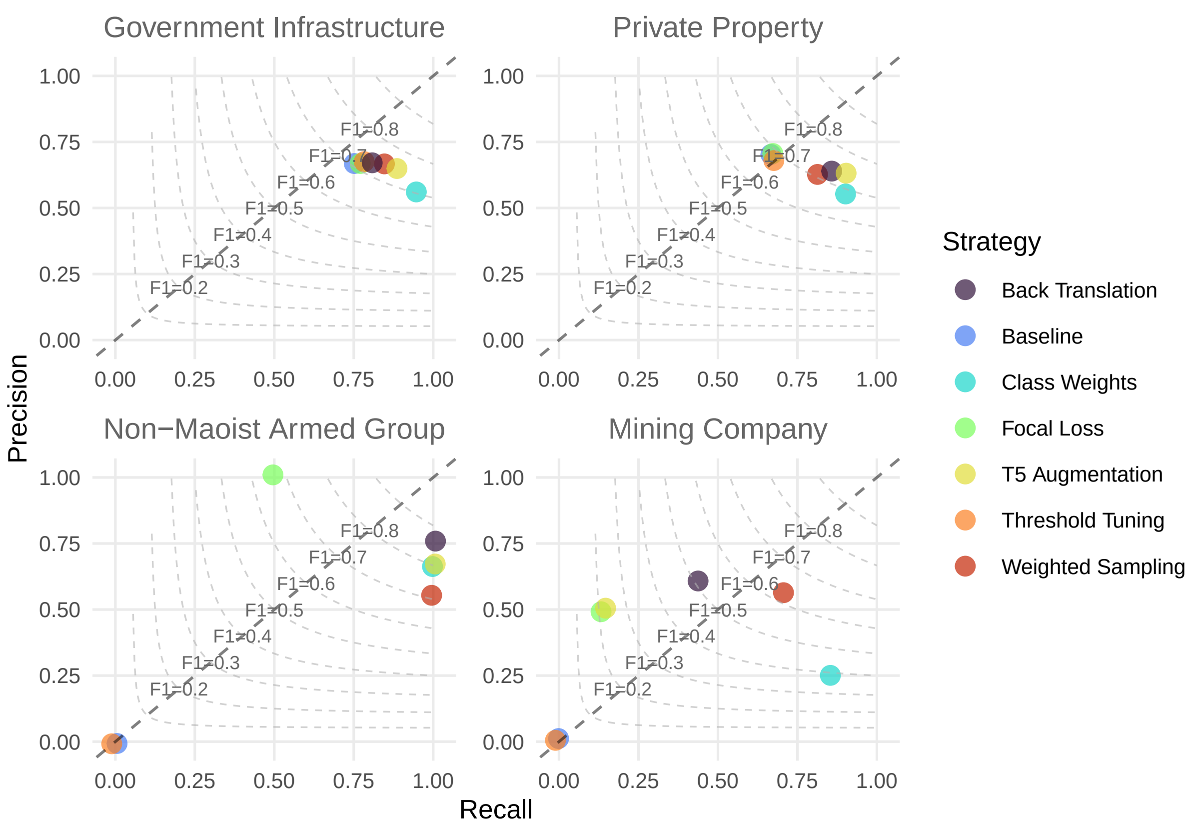


Figure 8: Precision-Recall Plots for Imbalance Handling Strategies

Back-translation augmentation consistently achieves the highest F1 scores (0.72-0.76) across rare categories through well-balanced precision-recall profiles. This balanced approach suggests that high-quality synthetic examples enable models to develop robust decision boundaries that perform reliably across both precision and recall dimensions.

Focal loss demonstrates a precision-specialist profile, garnering perfect precision (1.0) for "Non-Maoist Armed Group" classification while sacrificing recall (~0.5). This pattern aligns with focal loss's theoretical foundation of focusing learning on the hardest examples, and in doing so the model becomes highly conservative but accurate in its positive predictions. This precision-emphasis may be operationally valuable in applications where false positives carry high analytical costs.

Weighted sampling and class weights occupy intermediate positions, suggesting these approaches provide reasonable compromises between precision and recall without strongly favoring either dimension. T5 augmentation shows solid, consistent performance across categories, typically achieving good precision-recall balance albeit with slightly lower overall F1 scores than back-translation.

These precision-recall profiles have important implications for operational deployment. High-precision strategies like focal loss may be preferred when human analysts must review flagged events, as they minimize false positive rates that could overwhelm analytical capacity. Conversely, high-recall approaches may be more appropriate for comprehensive monitoring applications where missing true events carries greater costs than processing false positives. The balanced profiles achieved by augmentation strategies suggest they may be most suitable for general-purpose classification where both precision and recall matter equally.

Conclusion

This study evaluated six encoder-based transformer architectures and one generative GPT model on three classification tasks drawn from the South Asia Terrorism Portal: perpetrator attribution, action type, and target type. Across tasks, fine-tuned encoders achieved strong performance on high-frequency labels even at small data fractions, while low-frequency labels remained challenging until addressed with targeted imbalance-handling strategies. Back-translation through contextually relevant languages proved somewhat effective in classifying previously unlearnable categories.

The GPT-4.1 mini few-shot baselines add a practical reference point. On perpetrator classification, a single-label task with three well-separated categories, GPT-4.1 mini achieved competitive F1 with zero labeled training data. On action type it performed respectably, though fine-tuned encoders surpassed it once trained on roughly one-eighth of the available data. On target type, where the taxonomy is more granular and the label space larger, the few-shot generative approach struggled considerably. These results suggest that the amount of labor NLP tools can save depends heavily on task structure.

For researchers deciding how to invest their time, these findings imply a practical decision tree. When a classification task involves a small number of well-defined categories, a generative model with a carefully constructed prompt may be sufficient out of the box, requiring no labeled data at all. When the taxonomy is more complex or the generative baseline falls short, the results here show that fine-tuned encoders overtake GPT baselines with relatively modest annotation efforts, roughly on the order of one thousand labeled examples rather than tens of thousands. The traditional assumption that automated coding requires massive hand-coded training sets no longer holds. Even a targeted, limited annotation campaign can yield a classifier that outperforms both manual scaling and zero-shot generative approaches.

A related question that the field rarely confronts directly is what level of automated classification performance justifies replacing human coders. The implicit standard is that higher accuracy is always better, but human inter-coder reliability is itself imperfect. Major conflict datasets disagree at non-trivial rates on event classification for the same conflicts (Eck 2012). The appropriate benchmark for automated systems may not be perfection but parity with trained human coders on equivalent tasks. Researchers should consider what level of classification error is tolerable for their downstream analysis, particularly in light of evidence that treating predicted labels as error-free biases parameter estimates even when classification accuracy exceeds 90 percent (Egami et al. 2024).

Even with these methodological advances, adoption is not frictionless. Most political scientists are trained in inferential statistics and R, while the NLP tools evaluated here originate from data science and are implemented in Python. Generative AI lowers the entry barrier in that prompting a model requires no machine learning expertise, but fine-tuning workflows still carry a learning curve. Funding remains a necessity: for annotation (even if smaller-scale than previously assumed), for computational resources, and for methodological training. The history of abandoned or defunded coding projects illustrates what happens when event-data infrastructure depends on continuous grant support (Nardulli, Althaus, and Hayes 2015; Croicu and Weidmann 2015). Sustainable adoption requires not just better models but institutional commitments to maintaining them.

Researchers working with sensitive conflict data face an additional constraint. Commercial generative APIs require transmitting text, potentially containing information about informants, ongoing operations, or vulnerable populations, to third-party servers. Open-source encoder models that can be fine-tuned and deployed locally address these data-sovereignty concerns in ways that cloud-based generative services cannot (Weber and Reichardt 2023; Grossmann et al. 2023). This consideration may tip the build-versus-buy calculus toward fine-tuned models even when generative baselines achieve acceptable accuracy.

Looking ahead, the most promising direction may be hybrid pipelines that combine the strengths of both paradigms, employing generative models for rapid prototyping and initial labeling, followed by fine-tuned encoders with requisite imbalance handling for production deployment where speed, cost, privacy, or accuracy on complex taxonomies demand it. As both model families continue to improve, the practical question for conflict researchers is shifting from whether automated coding is viable to how best to integrate it into existing research workflows.

Appendix

A1. Decoder-Only LLM Baseline Results

Table 1 presents the full results of zero-shot and few-shot classification using three decoder-only LLMs: GPT-4o-mini, GPT-4.1 mini, and GPT-5.2. These models were evaluated on the same test sets used for the encoder-based models. GPT-4.1 mini few-shot achieved the best overall performance among the decoder-only models and is used as the baseline reference in Figures 1–5.

Table 1: Decoder-Only LLM Classification Results (Zero-Shot and Few-Shot)

Task	Model	Mode	Micro F1	Macro F1	Weighted F1
Perpetrator	GPT-4o-mini	Zero-shot	0.819	0.605	0.811
Perpetrator	GPT-4o-mini	Few-shot	0.829	0.680	0.837
Action Type	GPT-4o-mini	Zero-shot	0.857	0.825	0.867
Action Type	GPT-4o-mini	Few-shot	0.892	0.876	0.897
Target Type	GPT-4o-mini	Zero-shot	0.541	0.523	0.589
Target Type	GPT-4o-mini	Few-shot	0.515	0.508	0.541
Perpetrator	GPT-4.1-mini	Zero-shot	0.880	0.679	0.872
Perpetrator	GPT-4.1-mini	Few-shot	0.899	0.783	0.905
Action Type	GPT-4.1-mini	Zero-shot	0.893	0.882	0.895
Action Type	GPT-4.1-mini	Few-shot	0.896	0.878	0.898
Target Type	GPT-4.1-mini	Zero-shot	0.655	0.576	0.678
Target Type	GPT-4.1-mini	Few-shot	0.523	0.521	0.535
Perpetrator	GPT-5.2	Zero-shot	0.889	0.715	0.884
Perpetrator	GPT-5.2	Few-shot	0.883	0.759	0.882
Action Type	GPT-5.2	Zero-shot	0.893	0.870	0.898
Action Type	GPT-5.2	Few-shot	0.910	0.889	0.914
Target Type	GPT-5.2	Zero-shot	0.628	0.551	0.699
Target Type	GPT-5.2	Few-shot	0.627	0.560	0.653

A2. Few-Shot Prompts

For each task, the decoder-only models received a system prompt defining the classification categories and a user prompt presenting the incident text. In the few-shot condition, labeled examples were inserted as user–assistant turn pairs before the test instance. The full prompts and examples are reproduced below.

Perpetrator Task

System prompt:

You are classifying conflict events from India's Maoist insurgency. Given an event description, identify the perpetrator/action-taker.

Categories:

- Maoist: Maoist insurgents, Naxalites, or affiliated groups
- Security: Government security forces, police, CRPF, military
- Unknown: Cannot be determined from the text

User prompt template:

Event: {text}

Respond with only one word: Maoist, Security, or Unknown

Few-shot examples (3):

1. "Maoists killed Constable Krishnalal Dhurtlahare in Dantewada District." □ **Maoist**
2. "Acting on a tip off, Police arrested a CPI-Maoist cadre, identified as Bhola Ram, from Basata village under Telpa Police Station limits in Arwal District." □ **Security**
3. "Bokaro Police and the CRPF troopers were engaged in an encounter with the CPI-Maoist cadres at Kanshidih forest area in Jhumra under Gomia Police Station of Bokaro District." □ **Unknown**

Action Type Task

System prompt:

You are classifying conflict events. An event may involve multiple action types.

Categories:

- Armed Assault: Armed attacks, shootings, ambushes, firefights
- Arrest: Detention, capture, apprehension by authorities
- Bombing: Explosions, IEDs, landmines, blasts
- Infrastructure: Attacks on facilities, sabotage, arson of buildings
- Surrender: Voluntary surrender, laying down arms

- Seizure: Raids, confiscation of weapons/materials
- Abduction: Kidnapping, hostage-taking

User prompt template:

Event: {text}

List all applicable categories, separated by commas. If none apply clearly, respond "None".

Few-shot examples (5):

1. "CPI-Maoist cadres shot dead a leader of the Congress party, Shaik Sabakthulla, at Madithadu village in the Cuddapah District." □ **Armed Assault**
2. "Three Maoists surrendered before the Police in Dornapal town of Sukma District." □ **Surrender**
3. "Two CRPF troopers were killed and three others were injured when Maoists triggered an explosive, when they were on way to a polling station in Jamui District parliamentary constituency." □ **Bombing**
4. "Three cadres of the PLFI, a breakaway faction of the CPI-Maoist, including an 'area commander', were arrested from Dhodhtritoli under Raidih Police Station in Gumla District. Gumla SP Jatin Narwal said the arrested PLFI cadres have been identified as 'area commander' Durjan Singh, Surendra Pal Singh and Bandhan Oraon - all residents of the same Police Station areas of the District. 'Police recovered a country-made pistol, eight sim cards, a cell phone and four live bullets besides a dairy containing names of the persons from whom PLFI activists extorted levy,' said the SP." □ **Arrest, Seizure**
5. "Three abducted Policemen, identified as Lakhan Netam, Chandrasekhar Thakur and Ramprakahs Tiwari of Amabeda Police Station, were released by the CPI-Maoist in the Kanker District. The released Policemen said they were abducted when they had gone to Sode village under Amabeda Police Station to make telephone calls to their homes. The extremists also took their mobile handsets and two motor cycles. According to the Police, the Maoists had taken them deep into the forest, tied them to trees and threatened them to quit the Police service." □ **Abduction**

Target Type Task

System prompt:

You are classifying the targets of conflict events. An event may have multiple targets.

Categories:

- Maoist: Maoist insurgents or their camps/facilities
- Security: Police, military, paramilitary forces
- Civilians: Non-combatant individuals
- Government Officials: Politicians, bureaucrats, elected officials
- Government Infrastructure: Government buildings, roads, bridges, railways
- Private Property: Civilian homes, businesses, vehicles
- Mining Company: Mining operations, personnel, or equipment
- Non-Maoist Armed Group: Other armed groups (not Maoist or government)
- No Target: No specific target (e.g., surrenders)

User prompt template:

Event: {text}

List all applicable categories, separated by commas.

Few-shot examples (5):

1. "One CRPF trooper was injured when CPI-Maoist cadres triggered a series of landmines targeting the SF personnel on a combing operation in Rudakota area" □ **Security**
2. "A husband and wife duo, Durjodhan Rajkowar and his wife Lalita, members of a Maoist squad operating in the Ayodhya Hills of Purulia District surrendered before the District Police." □ **No Target**
3. "CPI-Maoist cadres blew up a newly constructed health sub-centre building in Serendag village under Herhanj Police Station in Latehar District." □ **Government Infrastructure**
4. "A civilian and 11 Policeman were killed and three sustained injuries in an ambush set by the CPI-Maoist cadres in Metlaperu village forests under Bhadrakali Police Station area of Bijapur District." □ **Civilians, Security**
5. "About 150 cadres of the CPI-Maoist raided a stone-mine in Bastar region for explosives. However, when they could not find any explosive, they set ablaze eight stone crusher machines at Partha and Darbha areas of the region." □ **Mining Company**

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